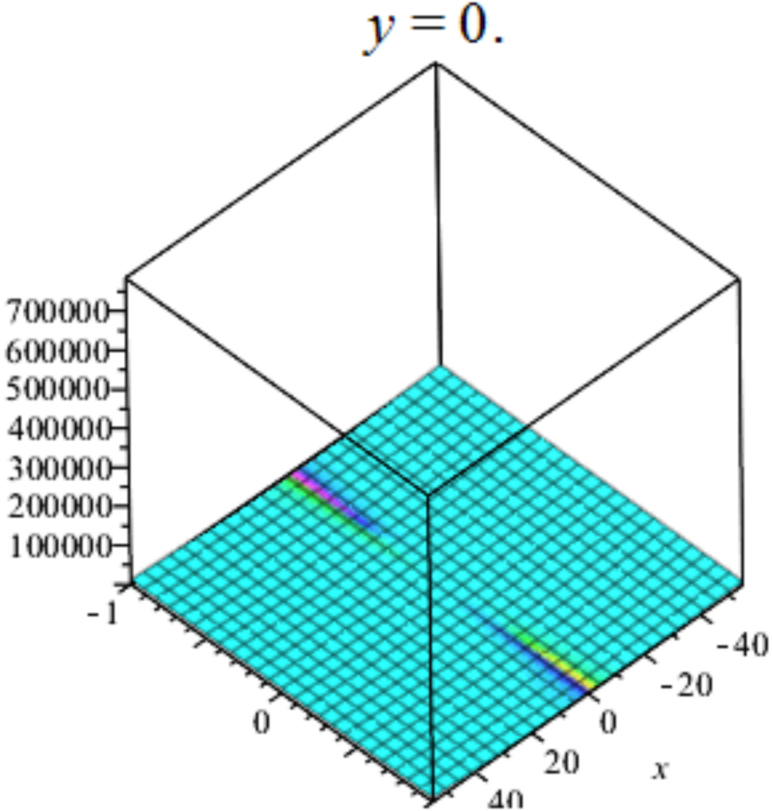
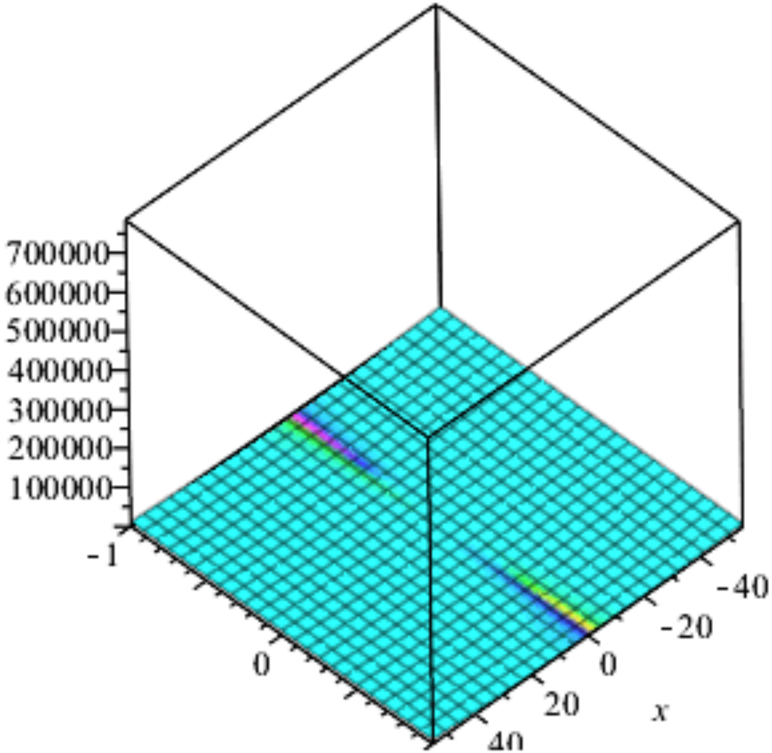
elegant connection to PDEs



Homogeneous Monge-Ampere equation

$y = 0.$

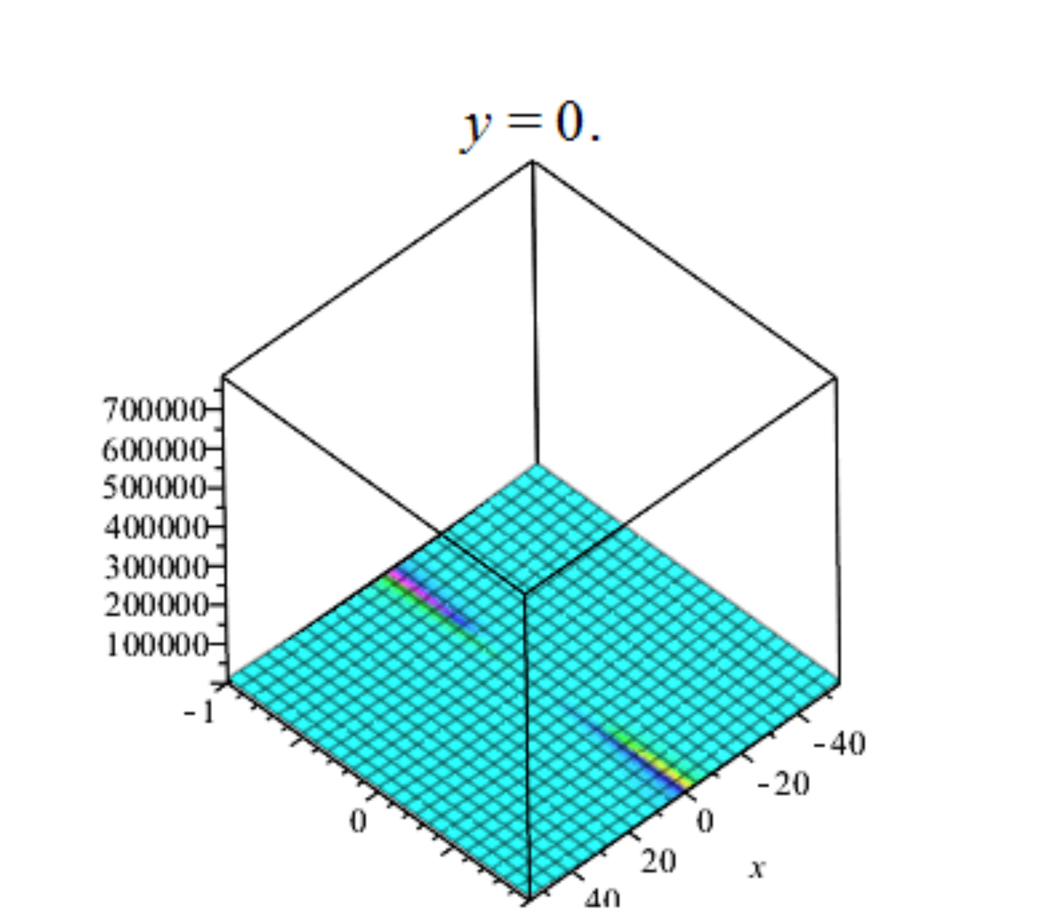


Homogeneous Monge-Ampere equation

Monge-Ampère Equation

- Brenier's theorem provides an **elegant connection to PDEs**
- Given boundary conditions, implies the existence of the OT map $T(x) = \nabla \varphi(x)$ between them
- This leads to a nonlinear PDE that φ must satisfy:

$$\det(D^2\varphi(x)) = \frac{\mu(x)}{\nu(\nabla \varphi(x))} \quad \text{(Monge-Ampère equation)}$$

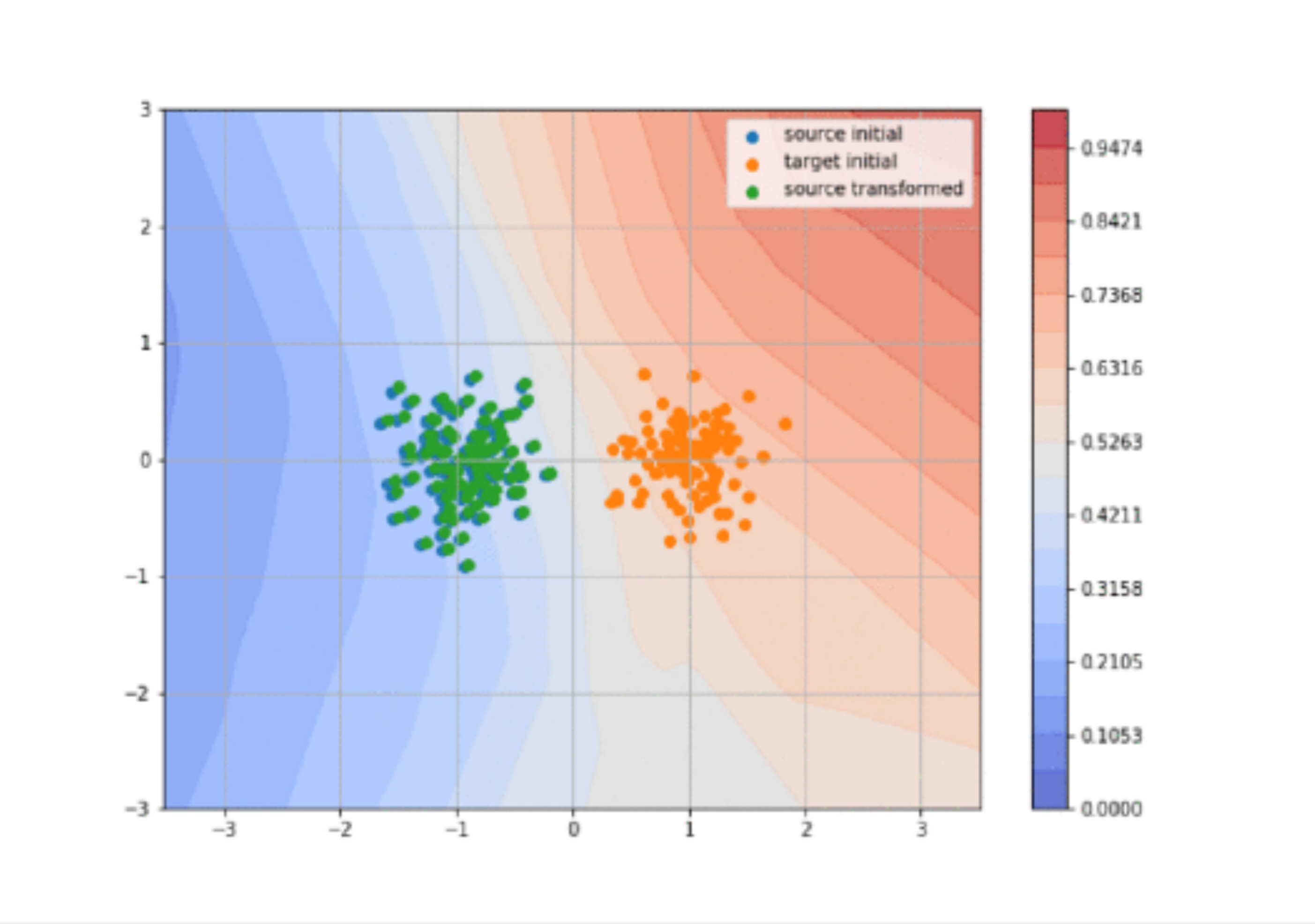


Homogeneous Monge-Ampere equation
10

- Appears in: control theory, optics, image registration, **neural OT**

Applications of Monge Maps

Domain Adaptation



(credit: Arthur Pesah)

$$T_\sharp \alpha = \beta$$